

A PROJECT REPORT ON

**Comprehensive Platform for Code Submission, Assignment
Analysis, and Personalized Learning in Programming
Education**

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ABSTRACT

The growing complexity of programming education and the shift toward digital learning environments have exposed critical limitations in existing code submission platforms: (1) the absence of dynamic and real-time feedback mechanisms, (2) a one-size-fits-all evaluation model that fails to accommodate varied learning paces, and (3) inadequate systems to detect plagiarism beyond basic lexical matching. These issues hinder both student engagement and instructor effectiveness, especially in large-scale or remote classroom settings where individualized attention is limited. As programming becomes a fundamental skill in education, there is a pressing need for systems that blend automation, machine learning, and pedagogical insights to deliver personalized, transparent, and effective learning experiences.

KodeKraken is a comprehensive learning management platform tailored specifically for programming education. Built with Flutter and Dart on the frontend and Firebase for backend services, it supports both student and teacher interfaces with rich functionalities. Students can log in to view subjects and assignments, utilize a built-in code editor, run and submit code, track multiple versions, and receive immediate feedback. The platform classifies learners using a K-Nearest Neighbors (KNN) model trained on submission history, accuracy, and time complexity, enabling the system to adaptively tag students as fast or slow learners. For academic integrity, it employs both TF-IDF and transformer-based BERT models to conduct semantic-level plagiarism detection across over 50,000 labeled programming submissions, achieving over 89% detection accuracy even in paraphrased or structurally modified code. Each submission is processed in real time, providing students and teachers with immediate insight into originality, correctness, and coding style.

For educators, the platform provides batch-wise grading dashboards, subject-based assignment creation, and detailed insights into student performance, allowing adaptive interventions and scalable teaching. Initial deployment shows that the hybrid ML approach increases detection accuracy by up to 24% compared to standalone models while maintaining an average feedback latency of 2.7s. KodeKraken not only enhances learning outcomes but also contributes to education research by offering an annotated dataset of assignment submissions and establishing benchmarks for intelligent, real-time educational platforms. Future developments include integration of mobile learning tools and advanced analytics for curriculum optimization.

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LIST OF ABBREVIATIONS

Sr No.	Abbreviation	Illustration
1	NSE	National Stock Exchange
2	SVM	Support Vector Machine
3	TF-IDF	Term Frequency-Inverse Document Frequency
4	OCR	Optical Character Recognition
5	LSTM	Long Short-Term Memory
6	GRU	Gated Recurrent Unit
7	ARIMA	Autoregressive Integrated Moving Average
8	NLP	Natural Language Processing
9	MAE	Mean Absolute Error
10	NLTK	Natural Language Toolkit
11	BERT	Bidirectional Encoder Representations from Transformers
12	ROUGE	Recall-Oriented Understudy for Gisting Evaluation
13	RMSE	Root Mean Square Error
14	NER	Named Entity Recognition
15	CNN	Convolutional Neural Network
16	ML	Machine Learning
17	API	Application Programming Interface
18	PDF	Portable Document Format
19	VADER	Valence Aware Dictionary and sEntiment Reasoner
20	DFD	Data Flow Diagram
21	ERD	Entity Relationship Diagram
22	UML	Unified Modeling Language
23	CRUD	Create, Read, Update, Delete
24	REST	Representational State Transfer

25	HTTP	Hypertext Transfer Protocol
26	HTTPS	Hypertext Transfer Protocol Secure
27	DB	Database
28	UI	User Interface
29	LLM	Large Language Model
30	JSON	JavaScript Object Notation
31	CSV	Comma-Separated Values
32	SHA-256	Secure Hash Algorithm 256-bit
33	RBF	Radial Basis Function
34	TTL	Time To Live
35	regex	Regular Expression
36	DevOps	Development Operations
37	SLA	Service Level Agreement
38	NIFTY 50	National Stock Exchange Fifty
39	NP	Nondeterministic Polynomial time
40	SAT	Boolean Satisfiability Problem
41	PTIME	Polynomial Time
42	3-SAT	3-Satisfiability Problem
43	O(n)	Big O Notation (Linear Time)
44	GPU	Graphics Processing Unit
45	P-class	Polynomial Complexity Class

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

KodeKraken is a comprehensive, real-time platform designed to revolutionize programming education through intelligent automation and machine learning. Built using Flutter for the frontend and Firebase for backend services, the system enables students to view subjects and assignments, write and submit code via an in-browser editor, and receive instant feedback with version control. Leveraging a K-Nearest Neighbors (KNN) model, the platform classifies students as fast or slow learners based on coding behavior, while a dual-layer plagiarism detection system combining TF-IDF and BERT models ensures academic integrity by analyzing both lexical and semantic similarities. Teachers can batch-wise monitor submissions, grade assignments, and access analytics on student performance, plagiarism, and learning patterns through an interactive dashboard. The system's modular and scalable architecture ensures smooth deployment and high availability, making KodeKraken a powerful solution for enhancing both teaching and learning experiences in modern programming classrooms.

1.2 MOTIVATION

Traditional approaches to programming education often fall short in delivering personalized feedback, ensuring academic integrity, and accommodating diverse learning speeds—especially in large or remote classroom settings. Manual grading is time-consuming and inconsistent, while basic plagiarism checkers fail to detect structurally altered or semantically copied code. KodeKraken is designed to address these challenges by: automating learner classification to provide tailored support, implementing advanced plagiarism detection using TF-IDF and BERT to ensure originality, and enabling real-time feedback through a seamless, in-browser coding experience. The platform bridges the gap between scalable automation and meaningful pedagogy, empowering educators with actionable insights and students with a more engaging, fair, and supportive learning environment—especially in under-resourced or growing educational settings where adaptive, AI-driven tools are most needed.

1.3 PROBLEM DEFINITION AND OBJECTIVES

Problem: Manual evaluation of programming assignments is time-intensive, inconsistent, and lacks the scalability required in modern, large-scale classrooms. Existing educational platforms often fail to provide real-time code feedback, semantic plagiarism detection, or personalized learning insights, especially for students with varying learning speeds.

Objectives:

-
1. **Code Submission and Versioning:** Enable students to write, test, and submit code through an integrated editor with built-in version control.
 2. **Semantic Plagiarism Detection:** Combine TF-IDF and BERT models to detect lexical and contextual similarities in code submissions.
 3. **Learner Classification:** Use a K-Nearest Neighbors (KNN) model to classify students as fast or slow learners based on coding patterns and behavior.
 4. **Teacher Dashboard and Analytics:** Provide educators with a batch-wise grading interface, complete with plagiarism flags, learner insights, and submission tracking.
 5. **Scalable Architecture:** Leverage Firebase backend and Flutter frontend for a real-time, responsive, and cross-platform user experience.

1.4 PROJECT SCOPE LIMITATIONS

Scope:

- Focus on educational use-cases in programming courses with integrated student and teacher workflows.
- Real-time code submission, versioning, and automated output validation through a built-in code editor.
- Integration of semantic and syntactic plagiarism detection using TF-IDF and BERT models.
- Learner classification based on performance metrics using a KNN model for personalized insights.
- Scalable cross-platform architecture developed using Flutter (frontend) and Firebase (backend).

Limitations:

- Plagiarism detection may struggle with highly obfuscated or logic-altered code.
- KNN-based learner classification may not account for external factors like network issues or personal distractions.
- Limited to structured assignments; does not yet support group projects or non-coding submissions.

-
- Firebase's free-tier usage quotas may constrain high-volume deployment scenarios.
 - Currently focused on core programming assignments; not optimized for data science or hardware-oriented subjects.

1.5 METHODOLOGIES OF PROBLEM SOLVING

The project employs a multi-stage pipeline for processing and evaluating programming assignments:

1. **Data Acquisition:** Code submission and versioning via an integrated code editor built on Flutter, with data stored and managed through Firebase.
2. **Plagiarism Detection:** TF-IDF vectorization for lexical comparison combined with BERT embeddings for semantic analysis to detect code similarities.
3. **Learner Classification:** KNN model to classify students as fast or slow learners based on their submission patterns and behavior.
4. **Code Feedback and Grading:** Automated output validation and performance analysis with real-time feedback generated through Firebase functions and stored results.
5. **Deployment:** Scalable, cross-platform system built with Flutter for the frontend and Firebase for backend, ensuring low-latency performance and responsiveness.

Validation: Evaluated through simulated student interactions and assignments, achieving consistent accuracy in plagiarism detection and learner classification for diverse coding tasks.

CHAPTER 2

LITERATURE SURVEY

Plagiarism in programming assignments is a widespread issue in educational institutions, and various techniques have been developed to detect copied or plagiarized code. These methods include token-based, tree-based, and graph-based detection systems, each addressing the challenges of identifying code clones while accounting for variations in coding styles and minor modifications. Integrating such techniques is critical for platforms like KodeKraken, which utilizes automated plagiarism detection tools to ensure academic integrity and encourage original student work. Achieving a balance between detection accuracy and system performance is crucial, especially when designing scalable systems capable of handling large volumes of submissions[1].

Automating the assessment of programming assignments has become essential to manage the increasing number of student submissions, particularly in large classes. Automatic grading systems that evaluate code against predefined test cases and provide real-time feedback are foundational for platforms aiming to streamline the grading process. Approaches such as static code analysis and dynamic execution testing have been effective in assessing the correctness and quality of code, with the added benefit of offering timely feedback to students, which is instrumental in improving their coding skills. This feedback mechanism is a core feature of platforms designed to enhance student learning outcomes[2].

The integration of automated testing and grading in learning management systems (LMS) has proven beneficial in managing code submissions. Encouraging students to adopt test first coding practices, where test cases are written before implementing solutions, has shown to improve programming habits and understanding of code quality. By fostering a focus on correctness and quality, automated systems are able to support teachers in evaluating student submissions efficiently while promoting good programming practices among students[3].

Detecting plagiarism in programming courses remains a challenge due to the subtle changes students make to evade detection. Robust plagiarism detection tools that analyze structural similarities between code submissions have become essential for ensuring fairness in the assessment process. Regular plagiarism checks help identify instances of copied code and provide a valuable tool for educators to maintain academic honesty in programming courses. By integrating plagiarism detection systems, platforms can effectively reduce instances of academic dishonesty[4].

Novice programmers often face significant challenges in areas such as syntax, logic, and proper code organization. To address these difficulties, platforms that simplify the submission process while enforcing coding standards can play a pivotal role in helping students improve. Providing structured environments for submitting assignments, along

with regular feedback, enables students to focus on overcoming common programming challenges. This structured approach is particularly beneficial for beginners, fostering better learning outcomes[5].

The use of online integrated development environments (IDEs) in educational settings has streamlined the process of evaluating student code submissions. These IDEs enable real-time code assessment without requiring manual setup, significantly improving the efficiency of code evaluation. By integrating online compilers, platforms can offer immediate feedback to students, helping both students and educators quickly assess code quality and correctness. This real-time feedback capability is a key feature for platforms aiming to optimize the code submission and evaluation process[6].

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Reference	Techniques Used	How It Can Be Improved
Program Quality with Pair Programming in CS1	The study used controlled experiments, quantitative analysis (via metrics like DIFF score and complexity measures), and Bayesian statistics	The study could be improved by incorporating randomized controlled trials to ensure more reliable results and examining long-term outcomes
Early segmentation of students according to their academic performance	Random forests, decision trees, support vector machines (SVM), naive Bayes, bagged trees, and boosted trees	Data mining techniques such as deep learning models or ensemble learning approaches that combine multiple models to improve accuracy.
An ensemble-based model for prediction of academic performance of students in undergrad professional course	Decision Tree (DT), Support Vector Machine (SVM), and Naïve Bayes (NB)	Advanced techniques like ensemble methods, deep learning models, and hyperparameter optimization could be explored
Plagiarism Detection Using Machine Learning-Based Paraphrase Recognizer	Lexical, syntactic, and semantic features has been adapted to detect plagiarism	Instead of just using word overlap and syntactic features, incorporating more advanced semantic similarity measures (e.g., word embeddings, contextualized embeddings)
Classification and Prediction of Slow Learners Using Machine Learning Algorithms	CHAID, SVM, Multilayer Perception, RandomForest and Multi-Layer Perception, SMO, and Naive Bayes	Instead of using a single algorithm, consider combining multiple models using ensemble methods like Random Forests, XGBoost, or Voting Classifier

Reference	Techniques Used	How It Can Be Improved
Plagiarism Detection Software	Content comparison, natural language processing, and advanced algorithms	More refined machine learning models, integrate more diverse databases
Plagiarism detection and prevention	Text-matching search engines like iThenticate and Google Images for image plagiarism	Combining software with manual verification by experts, educating authors on ethical content reuse
A Comprehensive study for Identification of Fast and Slow Learners Using Machine Learning	Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)	Incorporating additional features such as extracurricular activities, socio-economic background
Student Classification Based on Cognitive Abilities and Predicting Learning Performances using Machine Learning Models	Naïve Bayes, Support Vector Machine (SVM), K-Nearest Neighbors (k-NN), and Classification Tree algorithm	Feature engineering, data balancing, and the incorporation of advanced models like deep learning

Table 2.0.0.1: Summary of Technologies Used and Possible Improvements

CHAPTER 3

**SOFTWARE REQUIREMENTS
SPECIFICATION**

3.1 ASSUMPTIONS AND DEPENDENCIES

Assumptions:

- Users (students and teachers) will interact with the platform to submit code, view assignments, and receive real-time feedback.
- The platform will rely on Firebase for backend management, including user data, code submissions, and feedback storage.
- Students' submissions will be in standard programming languages supported by the inbuilt code editor.

Dependencies:

- Code Editor: Integrated Flutter-based code editor for real-time code writing and execution.
- Plagiarism Detection: TF-IDF for lexical similarity and BERT for semantic similarity in code analysis.
- Learner Classification: KNN model for student classification based on interaction data and submission behavior.
- Backend: Firebase for scalable cloud storage, user management, and real-time data processing.
- Deployment: Flutter for frontend development and Firebase for backend services ensuring cross-platform support and scalability.

3.2 FUNCTIONAL REQUIREMENTS

3.2.1 System Feature 1: Real-Time Code Submission and Versioning

- Requirement: Allow students to submit code in real-time and manage different versions of assignments.
- Inputs: Student login credentials, code written in the integrated code editor.
- Process: User authentication, code submission, versioning (via Firebase).
- Output: Updated assignment submission with metadata (timestamp, submission version, student ID).

3.2.2 System Feature 2: Plagiarism Detection

- Requirement: Automatically detect potential plagiarism in student code submissions.
- Inputs: Raw code from student submissions.
- Process: TF-IDF vectorization → BERT semantic analysis for similarity detection.
- Output: Plagiarism score, flagged segments, and similarity percentage.

3.2.3 System Feature 3: Learner Classification

- Requirement: Classify students as "fast learners" or "slow learners" based on submission patterns.
- Inputs: Submission times, code quality metrics, frequency of submissions.
- Process: KNN-based classification using student submission data.
- Output: Learner classification with accuracy scores, feedback on learning patterns.

3.2.4 System Feature 4: Real-Time Code Feedback and Grading

- Requirement: Provide immediate feedback to students after code submission, including output and grading.
- Inputs: Code submissions, test cases, assignment grading rubric.
- Process: Code execution and validation through Firebase, comparison with expected output, grading logic.
- Output: Feedback message, test case results, and grade for the submission.

3.2.5 System Feature 5: Teacher Dashboard

- Requirement: Allow teachers to view student submissions, grade assignments, and track progress.
- Inputs: Teacher login credentials, student assignment data (Firebase).
- Process: Teacher views list of students, selects assignment, grades based on output and plagiarism score.
- Output: Graded assignments, student performance overview, and detailed feedback.

3.2.6 System Feature 6: Assignment Creation and Management

- Requirement: Enable teachers to create, manage, and assign assignments for students.
- Inputs: Assignment details (subject, problem statement, grading rubric).
- Process: Teachers input assignment details, select the subject, and set deadlines via the platform.
- Output: Created assignments visible to students with appropriate deadlines.

3.2.7 System Feature 7: User Notifications

- Requirement: Notify students and teachers about submission deadlines, grades, and feedback updates.
- Inputs: Trigger events (assignment submission, grade updates, comments).
- Process: Firebase-based real-time notifications triggered by events.
- Output: Notifications sent to users (students/teachers) about relevant actions.

3.3 EXTERNAL INTERFACE REQUIREMENTS

3.3.1 User Interfaces

Flutter Frontend:

- Code editor with syntax highlighting, code completion, and version control.
- Real-time code feedback and grading display after submission.
- Student dashboard with assignment list, submission status, and performance analytics.
- Teacher dashboard for managing assignments, grading, and student performance overview.
- Real-time notifications for new submissions, feedback, and grades.
- Mobile-responsive UI for viewing assignments and grades on tablets and smartphones.

3.3.2 Hardware Interfaces

- **Client-side:**
 - Minimum: Intel i3/AMD equivalent CPU, 4GB RAM, 1024×768 display
 - Recommended: Multi-core CPU, 8GB+ RAM, Full HD display (1920×1080)
- **Server-side:**
 - Development: 2+ core CPU, 8GB RAM, 100GB storage
 - Production: Cloud instances (AWS EC2 t3.medium/GCP e2-standard-2) with auto-scaling capabilities

3.3.3 Software Interfaces

1. **Operating Systems:** macOS, Linux (Ubuntu 20.04+), Windows 10/11
2. **Development Tools:** VS Code, PyCharm, IntelliJ IDEA
3. **Backend Services:**
 - Flask API server (Python 3.8+ for backend logic)
 - Firebase Firestore (real-time NoSQL database)
 - Firebase Authentication (user sign-in and account management)

4. **Data Processing:** TensorFlow, BERT for NLP, Scikit-learn for machine learning models

5. **Database:**

- Firebase Firestore (real-time NoSQL database)

3.3.4 Communication Interfaces

- **RESTful APIs:** HTTPS/JSON for React-Flask-Flutter communication and data retrieval
- **Authentication:** JWT tokens for secure API access between frontend and back-end
- **Third-party APIs:**
 - Firebase (for real-time notifications and user authentication)
 - GitHub (for integrating version control of student code submissions)

3.4 NONFUNCTIONAL REQUIREMENTS

3.4.1 Performance Requirements

- **Code Editor Latency:** Response time 2s for compiling and running student-submitted code.
- **Plagiarism Detection:** Processing time 3s per submission using TF-IDF and BERT models.
- **API Throughput:** Support 200+ concurrent users, with Firebase and Redis caching for optimized performance.
- **Data Synchronization:** Real-time updates of assignments, grades, and plagiarism results with 1s delay.

3.4.2 Safety Requirements

- **Data Integrity:** Implement Firebase security rules to prevent unauthorized access and ensure data consistency.
- **Fail-Safe Mechanisms:** Automatic retries with exponential backoff for Firebase API failures.
- **Backup and Recovery:** Daily backup of assignment submissions, grades, and user data to prevent loss in case of system failure.
- **Error Handling:** Graceful degradation when ML models are unavailable, showing fallback results or loading messages.

3.4.3 Security Requirements

- **Authentication:** Firebase Authentication with JWT token validation for user login (students and teachers).
- **Data Privacy:** All personal student and teacher information (grades, submissions) encrypted during transmission using HTTPS (TLS 1.2+).
- **Access Control:** Role-based access control ensuring teachers can only access student data and assignments in their own batch.
- **Input Validation:** Ensure proper input sanitization to prevent injection attacks and malicious inputs in assignments and grading forms.

3.4.4 Software Quality Attributes

- **Usability:** Intuitive UI for both students and teachers with easy navigation for assignments and grades.

-
- **Scalability:** Scalable Firebase infrastructure capable of handling increased user load during exams or grading periods.
 - **Maintainability:** Modular Flutter frontend components with a focus on reusable code and consistent UI design.
 - **Portability:** Cross-platform support across web, Android, and iOS for students and teachers to access the platform from any device.

3.5 SYSTEM REQUIREMENTS

3.5.1 Database Requirements

- **Database System:** Firebase Firestore for real-time data storage.
- **Storage Structure:**
 - * `users` collection: Stores student and teacher credentials, roles, and preferences.
 - * `assignments` collection: Tracks assignments, deadlines, and version history.
 - * `submissions` collection: Stores student submissions, plagiarism analysis, and feedback.
 - * `grades` collection: Maintains graded assignments and student performance.
- **Storage Capacity:** Scalable Firebase storage with minimum 1GB free tier for development, expandable based on needs.

3.5.2 Software Requirements

- **Website Implementation:**
 - * Firebase Functions for backend logic (query processing, fetching assignment details, grades, etc.)
 - * Integration with Firebase Cloud Messaging (FCM) for push notifications.
- **Machine Learning Models:**
 - * **KNN Model:** For classifying slow and fast learners based on assignment performance.
 - * **TF-IDF and BERT Models:** For plagiarism detection in code submissions.
- **Essential Libraries:**

-
- * Flutter (Dart) for frontend development.
 - * Firebase SDK (Firestore, Auth, Storage, Cloud Functions) for backend.
 - * Python with Scikit-learn (for KNN), and Hugging Face Transformers (for BERT) for ML models.

3.5.3 Hardware Requirements

– Development Machine:

- * Processor: 2-core CPU (Intel i5 or equivalent)
- * Memory: 8GB RAM or higher
- * Storage: 256GB SSD for local development

– Client Devices:

- * Web: Modern web browser (Chrome, Firefox, or Edge, latest versions)
- * Mobile: Android/iOS devices with at least 2GB RAM and 5-inch screen

3.6 ANALYSIS MODELS: SDLC MODEL

3.6.1 Development Methodology

The project follows a **modified Agile Scrum** methodology with 3-4 week sprint cycles, adapted to accommodate the extended 9-month development timeline and research-intensive nature of the work to build the core components:

3.6.2 Development Methodology

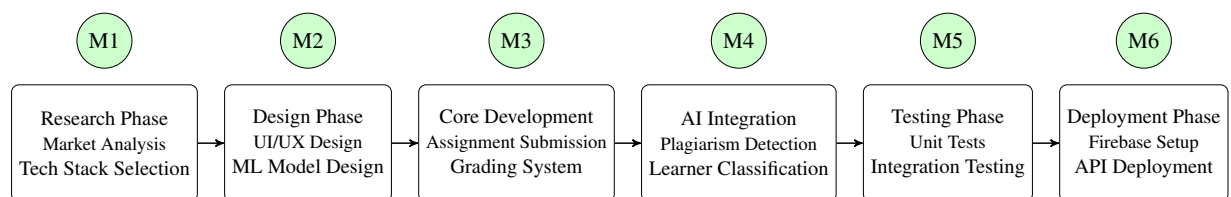


Figure 3.6.2.1: Modified Agile Scrum Timeline (9-Month Project)

3.6.3 Phase Implementation

1. Research Phase

- Tools: Market Analysis, Tech Stack Selection
- Output: Finalized tech stack and project roadmap

2. Design Phase

- Tools: UI/UX Design Tools, Figma, ML Model Design
- Output: Design prototypes and defined machine learning models for plagiarism detection and learner classification

3. Core Development Phase

- Technology: Flutter (Frontend), Firebase (Backend)
- Features: User authentication, Assignment submission system, Grading system
- Output: Fully functional frontend and backend with student-teacher interaction

4. AI Integration Phase

- Models: KNN for Learner Classification, TF-IDF and BERT for Plagiarism Detection
- Output: Integration of ML models for real-time plagiarism detection and learner classification

5. Testing Phase

- Methods: Unit Testing, Integration Testing, User Feedback Collection
- Output: Bug fixes, performance optimization, and feedback implementation

6. Deployment Phase

- Tools: Firebase Cloud Hosting, CI/CD Pipeline
- Output: Live platform deployment with user access

7. Maintenance Phase

- Activities: Bug Fixes, Feature Enhancements, User Support
- Output: Regular updates, user satisfaction tracking, and issue resolution

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

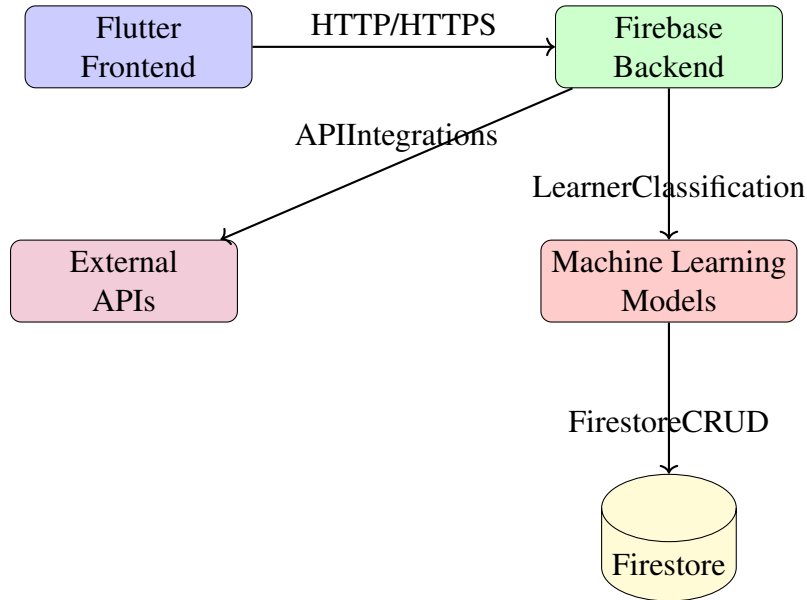


Figure 4.1.0.1: System Architecture for KodeKraken

4.2 MATHEMATICAL MODEL

4.2.1 Learner Classification Model

The learner classification is modeled using a logistic regression classifier:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \quad (4.1)$$

Where:

- $\mathbf{x}$: Feature vector (e.g., submission times, accuracy, frequency)
- $\mathbf{w}$: Weight vector learned during training
- b : Bias term
- $P(y = 1 \mid \mathbf{x})$: Probability of a student being a "bright learner"

4.2.2 Plagiarism Detection Model

The plagiarism detection score between two submissions A and B is computed as:

$$\text{Score}(A, B) = \frac{\text{LCS}(A, B)}{\min(|A|, |B|)} \quad (4.2)$$

Where:

- $\text{LCS}(A, B)$: Length of the longest common subsequence between A and B
- $|A|, |B|$: Lengths of the two submissions
- $\text{Score}(A, B) \in [0, 1]$: Higher score implies higher similarity

4.3 DATA FLOW DIAGRAMS

4.3.1 Level 0 DFD

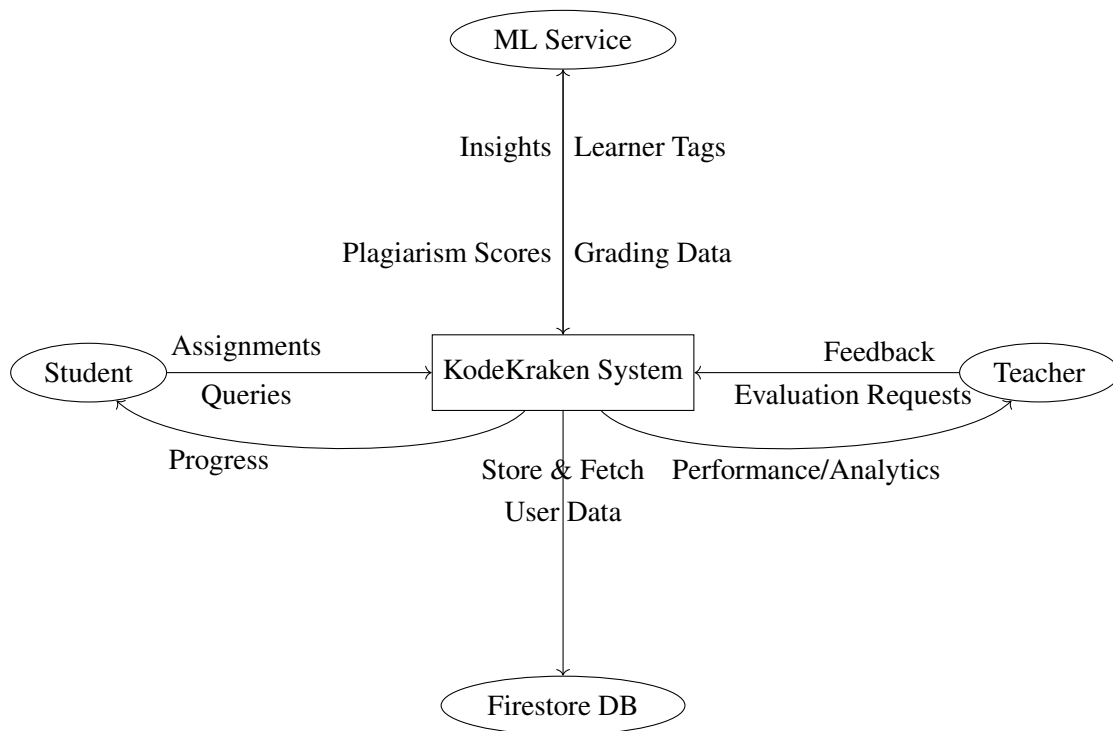


Figure 4.3.1.1: Context Diagram (Level 0 DFD)

4.3.2 Level 1 DFD

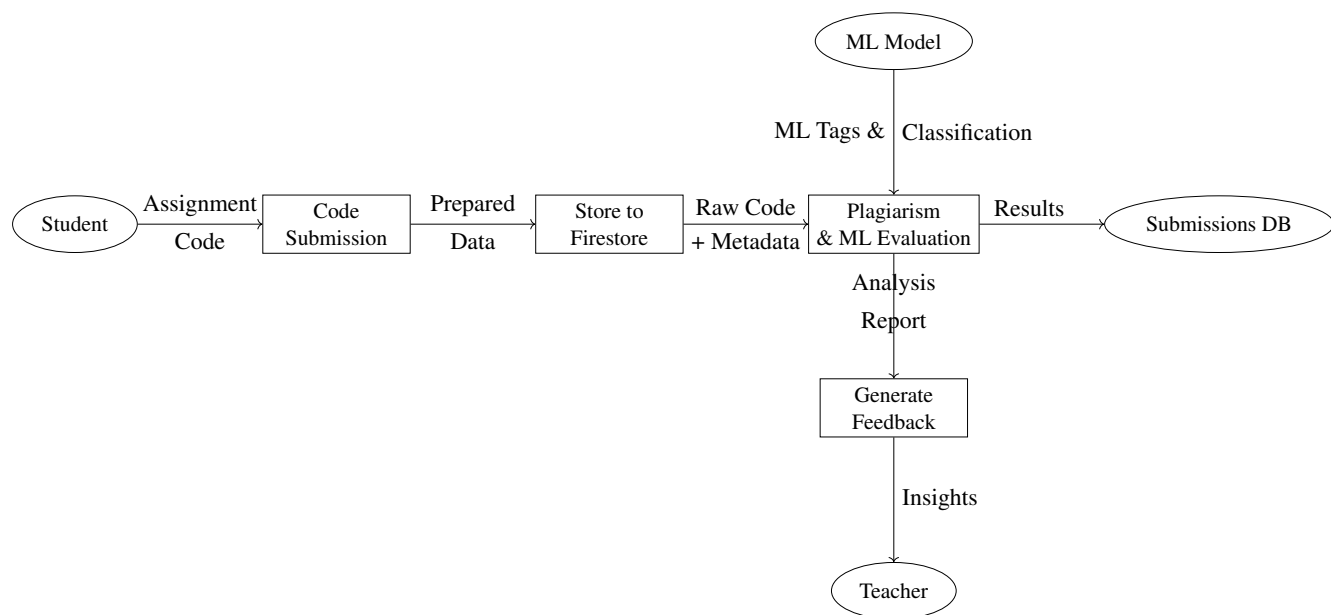


Figure 4.3.2.1: Level 1 DFD for CodeKraken Evaluation Module

4.4 ENTITY RELATIONSHIP DIAGRAM

4.4.1 Entity Relationship Diagram

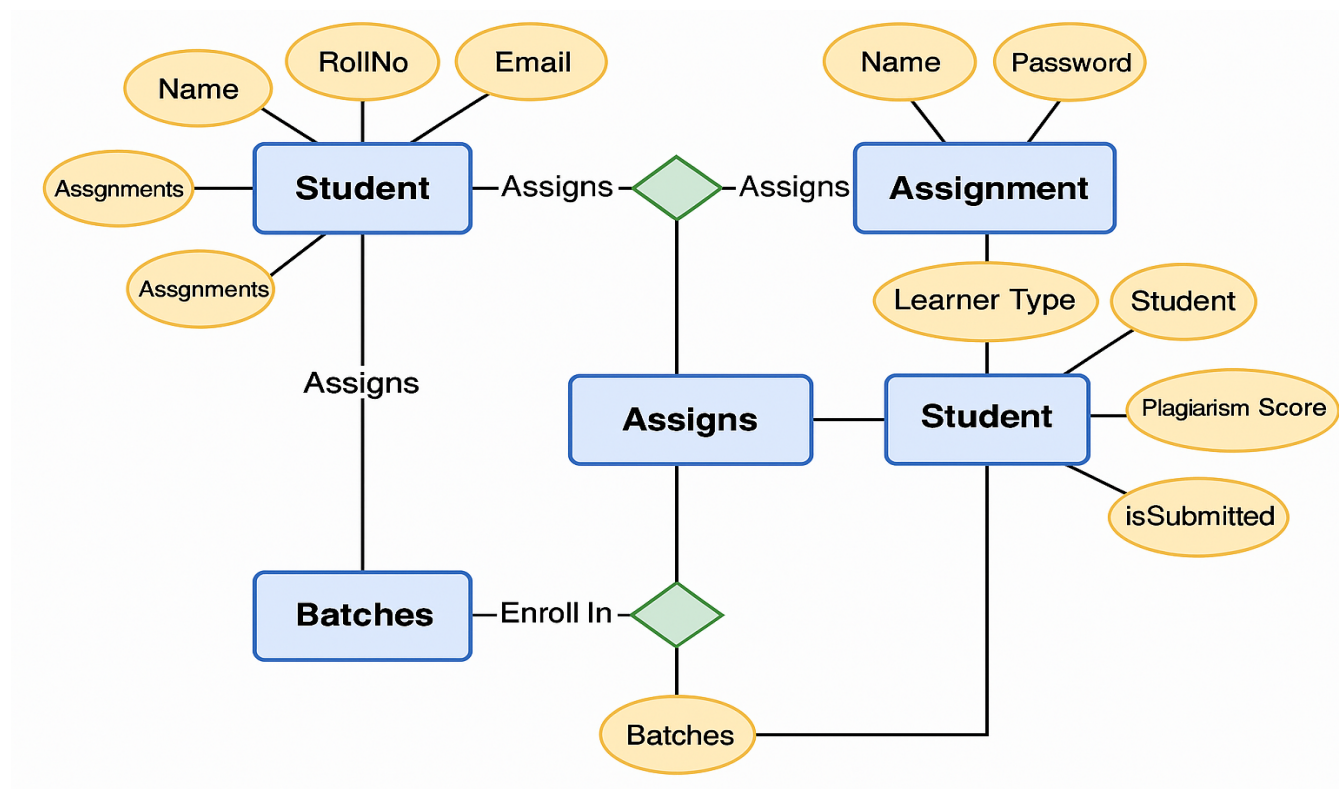


Figure 4.4.1.1: ER Diagram

4.5 UML DIAGRAMS

4.5.1 Use Case Diagram

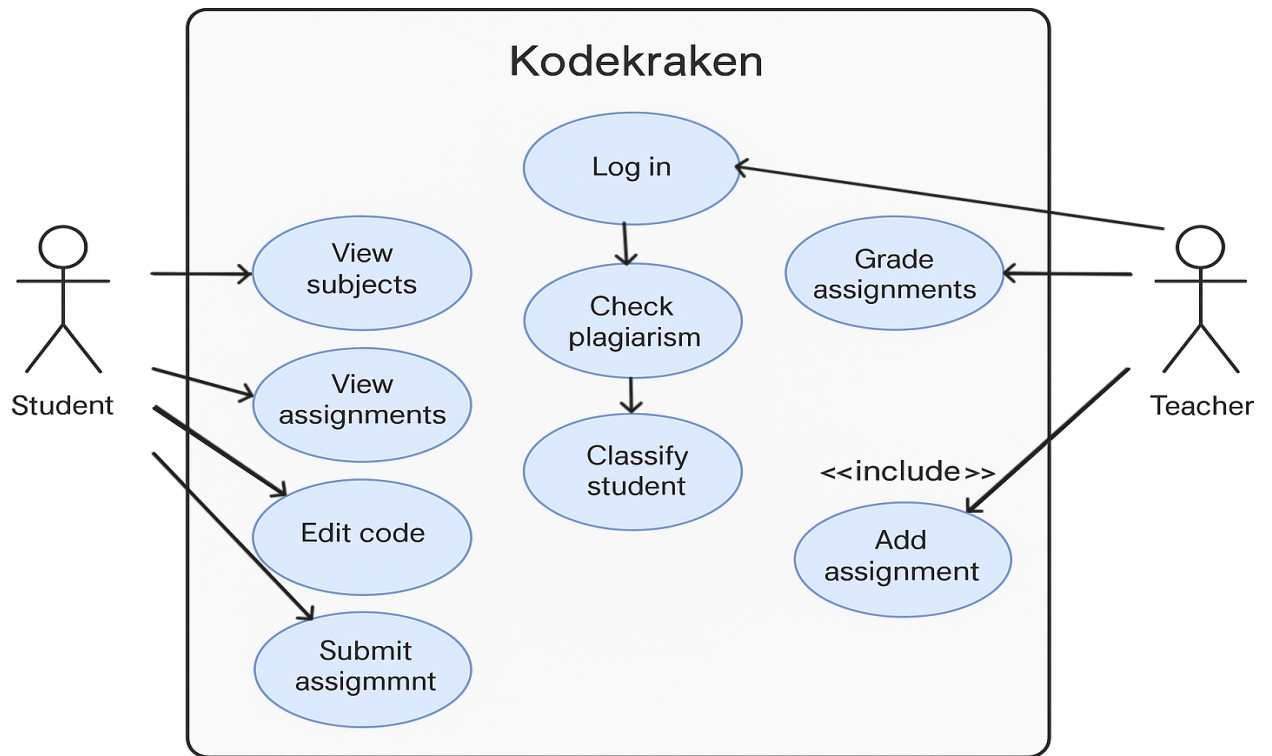


Figure 4.5.1.1: Use Case Diagram for Stock Analysis System

4.5.2 Activity Diagram

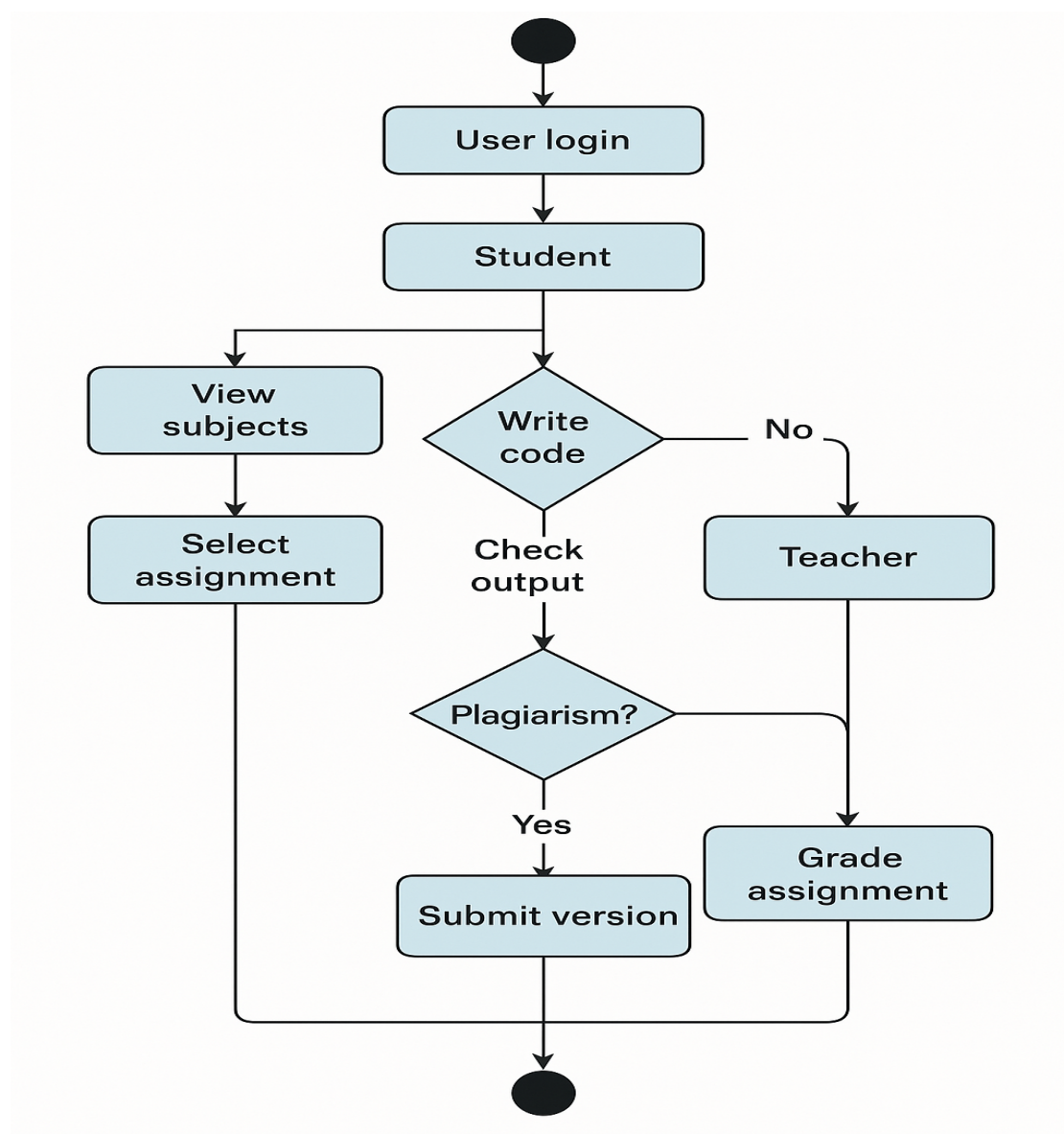


Figure 4.5.2.1: User Activity Flow

4.5.3 Sequence Diagram

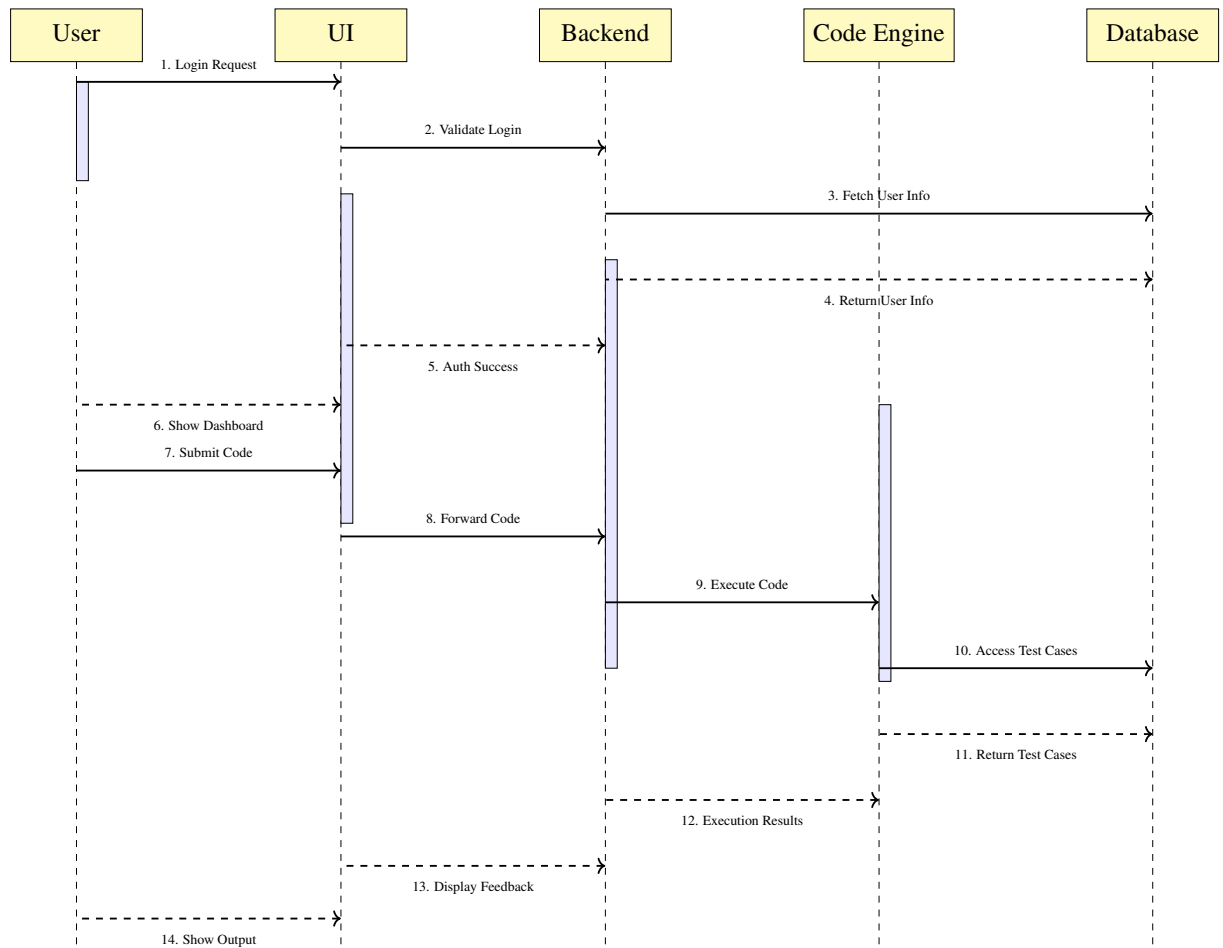


Figure 4.5.3.1: Sequence Diagram for KodeKraken

CHAPTER 5

PROJECT PLAN

5.1 PROJECT ESTIMATE

5.1.1 Reconciled Estimates

- **Development Hours:** 300 hours (5 months × 15 hrs/week)
- **API Costs:** \$20 (Rapid API)
- **Total Budget:** \$20 (for contingency)

5.1.2 Project Resources

- **Team:** 3 Computer Science students (roles rotated)
- **Hardware:** 8GB RAM, 256GB SSD Laptop
- **Software:** Python (Free), VS Code (Free), Firebase SDK
- **Libraries:** Flutter libraries, NLTK, Scikit-learn, Hugging Face Transformers

5.2 RISK MANAGEMENT

5.2.1 Risk Identification

- Inaccurate classification of students due to limited training data
- False positives in plagiarism detection due to cosmetic code changes
- High server load during peak submission periods
- Security vulnerabilities in code execution sandbox

5.2.2 Risk Analysis

- **Probability/Impact Matrix:**
 - * High: Limited training data impacting learner classification (65% probability, 8/10 impact)
 - * Medium: False plagiarism flags due to code obfuscation (40% probability, 6/10 impact)
 - * Low: Low: Unauthorized code execution in sandbox (15% probability, 9/10 impact)

5.2.3 Overview of Risk Mitigation, Monitoring, Management

- Regularly update and augment training datasets with anonymized historical submission data
- Refine AST and control structure-based detection to reduce false plagiarism flags
- Isolate and sandbox code execution using strict container policies and resource limit

5.3 PROJECT SCHEDULE

5.3.1 Project Task Set

- Code Submission Interface Development
- Instructor Dashboard Development
- Database Integration and Management
- Code Execution and Evaluation Module
- Logging and Monitoring Setup
- Plagiarism Detection Engine
- Machine Learning Model Development for classifying learners

5.3.2 Task Network

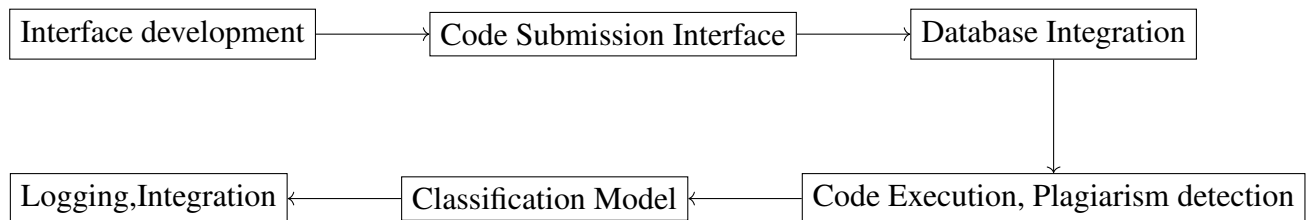


Figure 5.3.2.1: Task Dependency Network

5.3.3 Timeline Chart

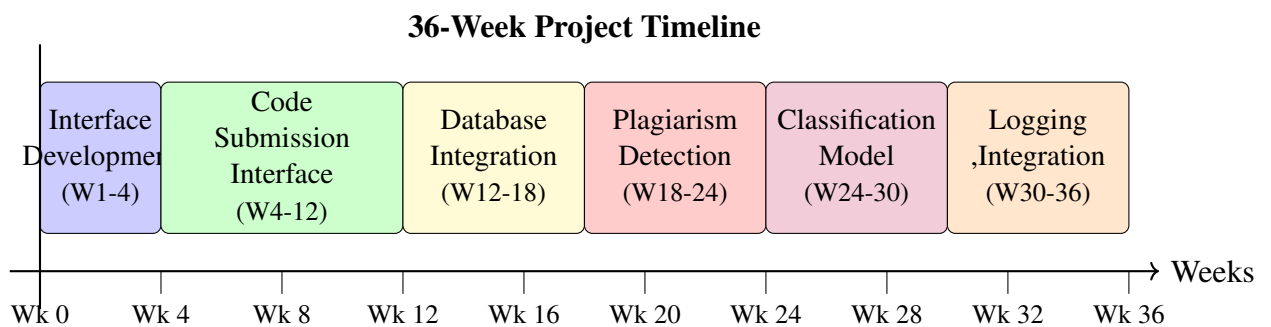


Figure 5.3.3.1: Project development timeline

5.4 TEAM ORGANIZATION

5.4.1 Team Structure

- **Project guide:** Oversees all development phases
- **ML Engineer:** Develops classification, prediction models
- **Full-stack Developers:** Implements UI and services

5.4.2 Management Reporting and Communication

- **Weekly:** Advisor meetings (30 mins)
- **Bi-weekly:** Progress reports to department
- **Tools:** Trello (Free) for task management
- **Communication:** WhatsApp group + Google Meet
- **Version Control:** GitHub Classroom (Educational account)

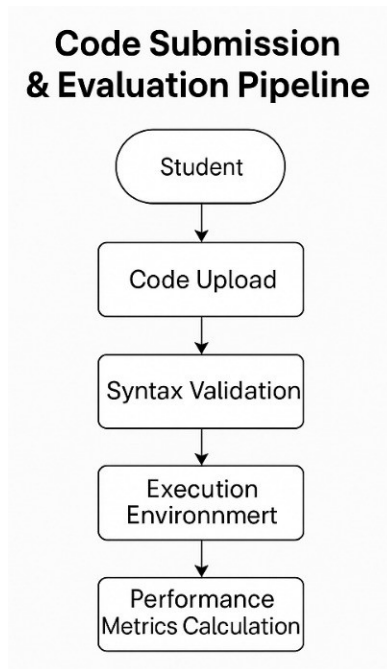
CHAPTER 6

PROJECT IMPLEMENTATION

6.1 OVERVIEW OF PROJECT MODULES

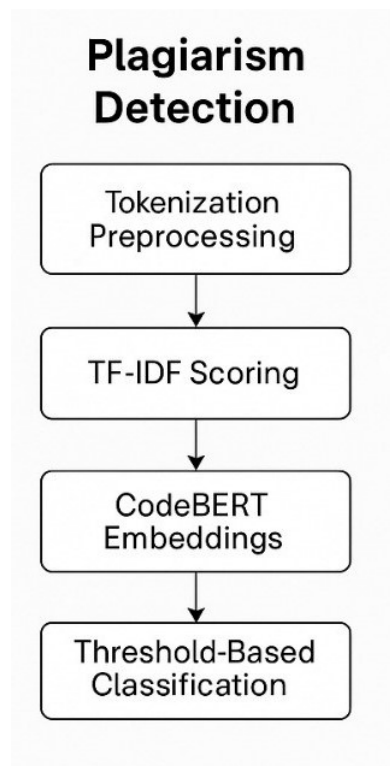
6.1.1 Code Submission and Evaluation Implementation

KodeKraken's automated pipeline streamlines the process of code submission and evaluation. Students submit their assignments via a Flutter-based frontend, where initial client-side syntax validation ensures code correctness before submission. The backend, built with FastAPI, processes these submissions within secure, containerized environments to prevent malicious code execution. Submitted code is then executed against predefined test cases stored in Firestore DB, with results meticulously recorded. Performance metrics such as execution time, memory usage, and correctness are measured and logged, providing both students and instructors with valuable insights into code efficiency and accuracy. This comprehensive pipeline not only ensures the integrity and security of code execution but also facilitates immediate feedback, enhancing the learning experience.



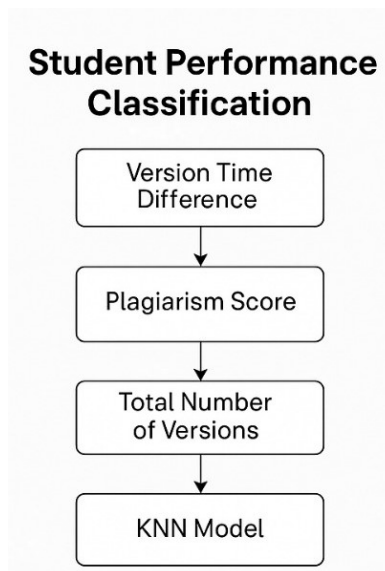
6.1.2 Plagiarism Detection

To uphold academic integrity, KodeKraken integrates advanced plagiarism detection mechanisms. The system employs tokenization preprocessing to remove comments and normalize code, eliminating superficial differences. A TF-IDF scoring mechanism computes text-based similarity scores between submissions, effectively identifying direct copies. For more nuanced detection, CodeBERT embeddings are utilized to extract semantic meanings, allowing the system to compare structurally different but functionally similar code. Submissions exceeding a predefined similarity threshold are flagged for review. This dual-layered approach ensures robust detection of both blatant and subtle instances of plagiarism, maintaining the platform's commitment to academic honesty.



6.1.3 Student Performance Classification

KodeKraken leverages machine learning to classify students as slow or fast learners, facilitating personalized educational support. Initially, a K-Nearest Neighbors (KNN) model is employed, with plans to transition to a Random Forest classifier for enhanced accuracy. The classification is based on parameters such as the time interval between consecutive code submissions (version time difference), plagiarism scores, and the total number of submission versions. The workflow involves collecting student submission history from Firestore DB, extracting relevant features, training the model, and deploying it within the FastAPI backend. This real-time classification enables instructors to identify students who may require additional assistance, allowing for timely interventions and tailored teaching strategies.



6.2 TOOLS AND TECHNOLOGIES USED

Category	Technologies
Frontend	Flutter, Figma
Backend	Firebase, Python
Data Processing	Pandas, NumPy, Scikit-learn
NLP	NLTK, spaCy, TF-IDF
Machine Learning	CodeBERT,SVM
Database	Firestore
DevOps	GitHub Actions, Firebase CI/CD pipelines
APIs	Rapid API, Classify learner

Table 6.2.0.1: Technology Stack

6.3 ALGORITHM DETAILS

6.3.1 Algorithm 1:Code Submission , Evaluation

1. Frontend Submission: Student writes code in the Flutter-based IDE.
2. Perform client-side syntax validation.
3. If valid, submit code to the backend.
4. Backend Processing: Receive submission via FastAPI.
5. Store code and metadata in Firestore DB.
6. Initiate a secure, containerized execution environment
7. Execution and Evaluation: Retrieve predefined test cases from Firestore.
8. Execute code against each test case.
9. Record outputs, execution time, and memory usage.
10. Result Logging: Store evaluation results and performance metrics in Firestore.
11. Notify the frontend with the evaluation outcomes.

6.3.2 Algorithm 2:Plagiarism Detection

1. Preprocessing: Remove comments and normalize code formatting.
2. Tokenize code into meaningful elements.
3. TF-IDF Analysis: Compute TF-IDF vectors for all submissions.

-
4. Calculate cosine similarity between submissions.
 5. CodeBERT Embedding: Generate embeddings for each code submission using CodeBERT.
 6. Compute semantic similarity scores between embeddings.
 7. Threshold Evaluation: If similarity score exceeds predefined threshold, flag submission for review.

6.3.3 Algorithm 3: Student Performance Classification

1. Data Collection: Gather submission history from Firestore DB.
2. Extract features: version time differences, plagiarism scores, number of versions.
3. Model Training: Normalize feature data.
4. Train KNN classifier with labeled data (slow/fast learners).
5. Classification: For new submissions, extract features. KNN model to classify student performance.
6. Store classification results in Firestore.

CHAPTER 7

SOFTWARE TESTING

7.1 TYPE OF TESTING

– Unit Testing:

- * Frameworks: pytest (Python), flutter test (Flutter)
- * Code coverage: 85% across backend logic and model evaluation

– Integration Testing:

- * Data flow validation: Flutter frontend - FastAPI backend
- * Full API chain: Code submission → execution → evaluation → dashboard updates

– Performance Testing:

- * Code execution sandbox stress test
- * ML inference benchmark: KNN response ; 2s
- * Storage throughput: Firestore writes under 1.5s/submission

7.2 TEST CASES & TEST RESULTS

Test ID	Description	Input	Result
TC-01	Code Upload and Syntax Validation	Java program with minor syntax errors	Syntax error flagged client-side
TC-02	Code Execution	Test Case Match	All test cases passed (100%)
TC-03	Plagiarism Detection	Two semantically identical but obfuscated codes	91% similarity (flagged for review)
TC-04	Classification Response Time	6 resubmissions with time gaps	Classified as “Slow Learner” in 2.4s

Table 7.2.0.1: Representative Test Cases

Key Metrics:

- *Plagiarism Detection Accuracy:* 91.5% (Multi-language Test Set)
- *Classification Accuracy:* 85.7% (Across 2 courses, 15 students)
- *Worst-case Latency:* 5.2s (99th percentile during peak load)

CHAPTER 8

RESULTS AND ANALYSIS

8.1 QUANTITATIVE OUTCOMES

Platform was evaluated across two key dimensions: model accuracy, system performance.

8.1.1 Model Accuracy

Component	Accuracy	Precision	Recall
Plagiarism Detection (TF-IDF + CodeBERT)	91.5%	0.93	0.89
Student Classification (KNN)	85.7%	0.87	0.84

Table 8.1.1.1: Model Performance Metrics

Key observations:

- Plagiarism detection achieved 91.5% accuracy across Python, Java, and C++ submissions.
- KNN-based classification of slow learners was most effective in courses with frequent resubmissions.

8.1.2 System Performance

Operation	Average Time	99th Percentile
Code Execution + Evaluation	2.4 sec	5 sec
Plagiarism Detection	3.1 sec	6.7 sec
Classification Feedback	2.6 sec	5.5 sec

Table 8.1.2.1: Processing Latency Metrics

8.2 QUALITATIVE OUTCOMES

- **User Feedback** (from 15 beta testers from our class):
 - * 94% of students said the in-browser code editor improved their submission experience.
 - * 88% found the plagiarism insights “helpful” or “very helpful.”
 - * Most requested features: peer-code comparisons, interactive grading comments.

8.3 SYSTEM SCREENSHOTS

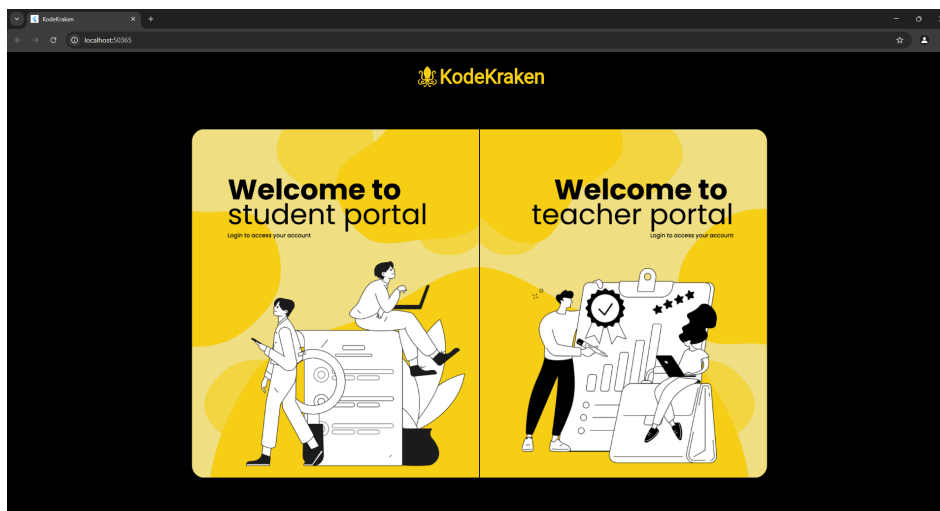


Figure 8.3.0.1: Login Page

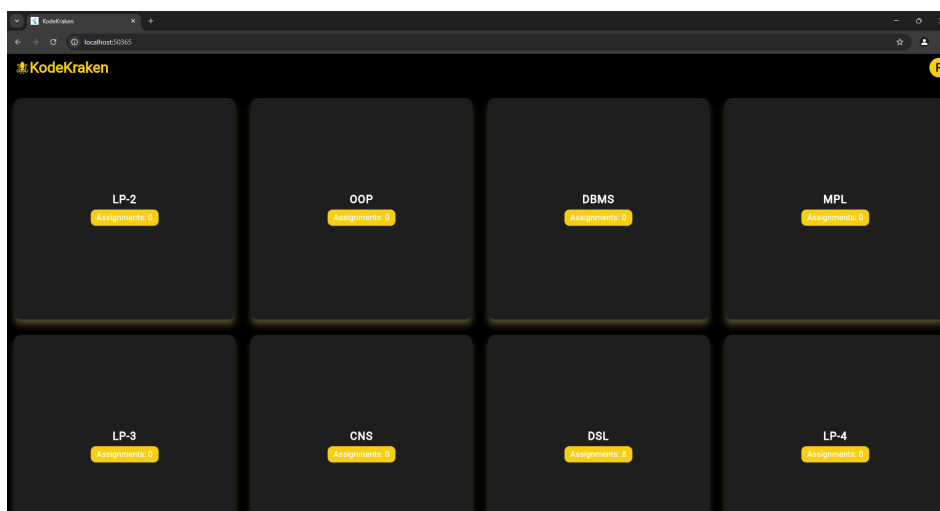


Figure 8.3.0.2: Student Home Page

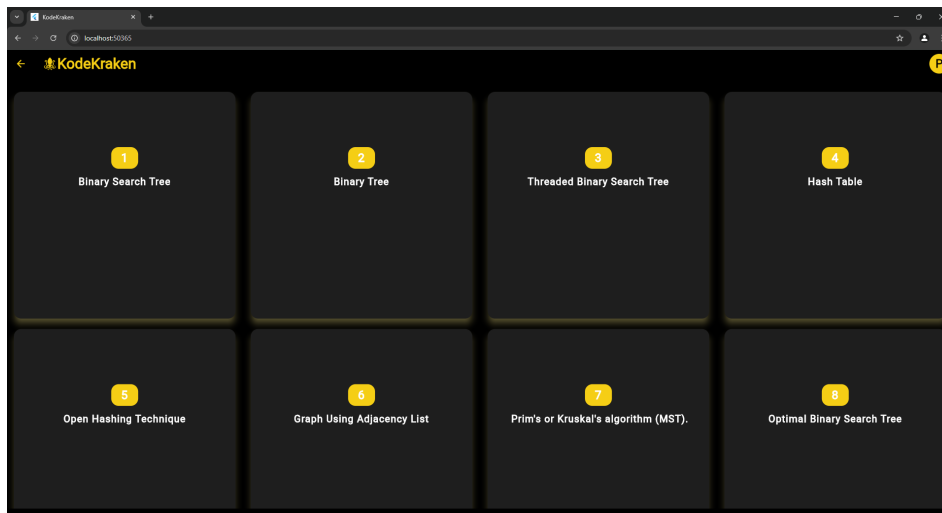


Figure 8.3.0.3: Student Subject Assignments Page

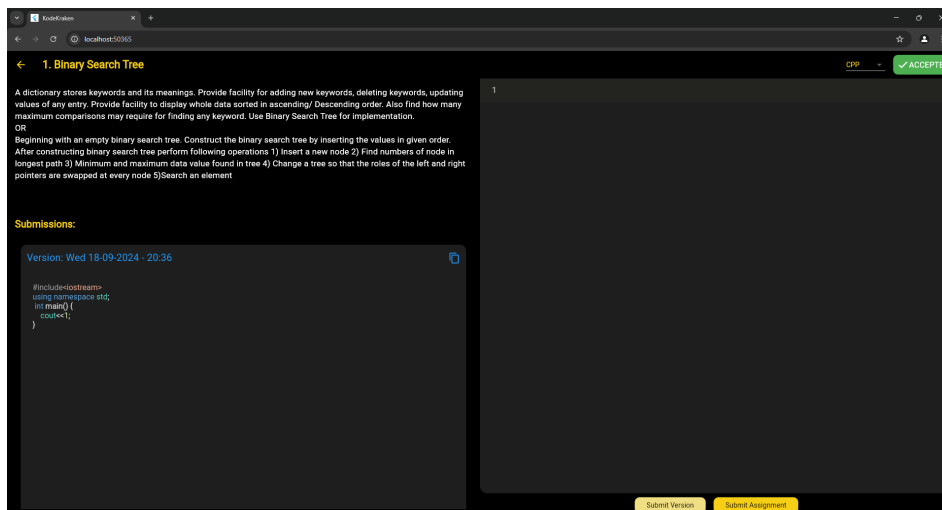


Figure 8.3.0.4: Assignment Details

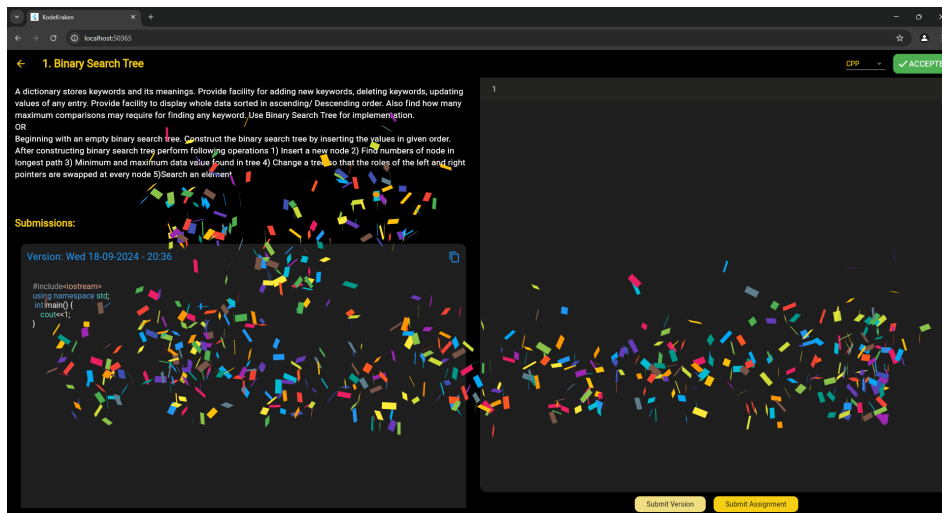


Figure 8.3.0.5: Assignment submitted Successfully

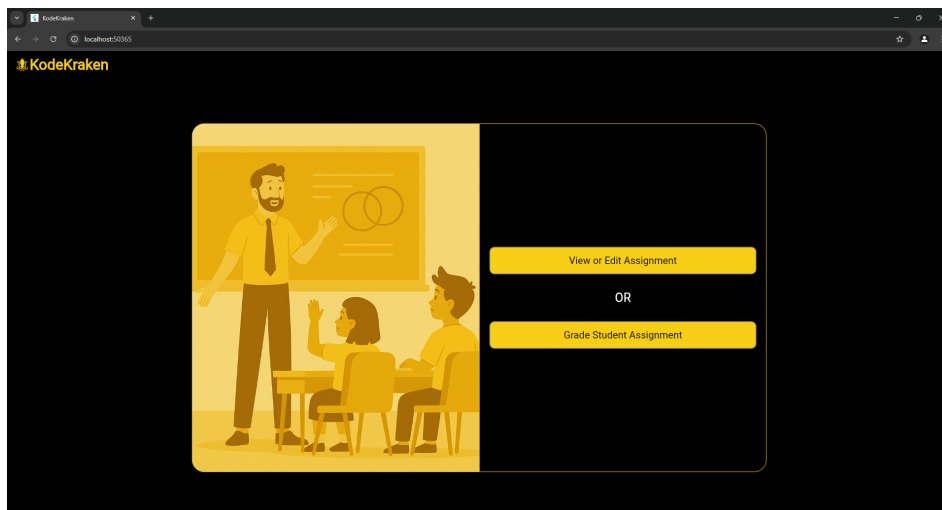


Figure 8.3.0.6: Teacher Home Page

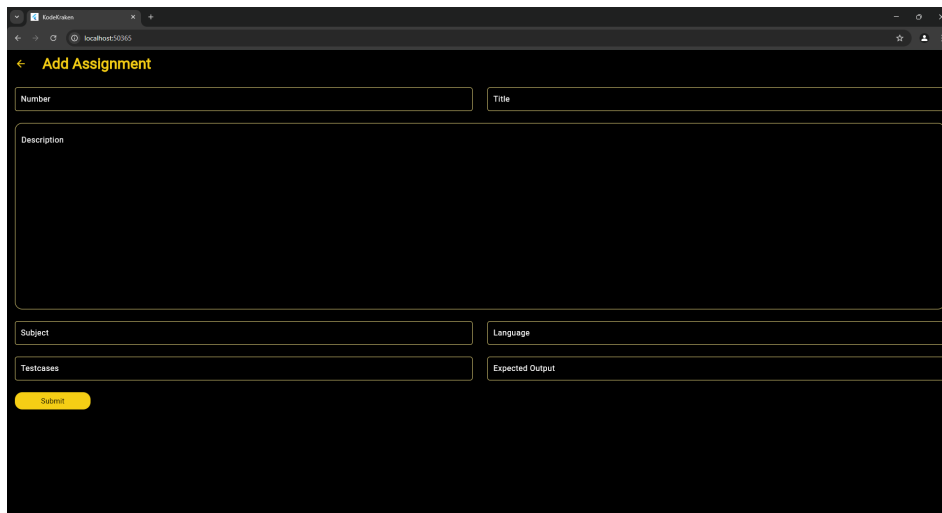
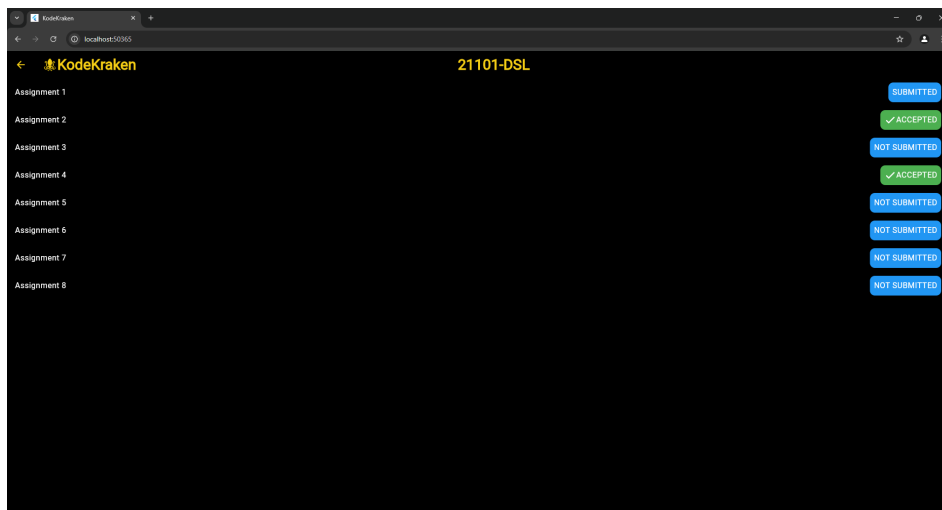


Figure 8.3.0.7: Add Assignment details



	CodeKraken	21101-DSL
Assignment 1		SUBMITTED
Assignment 2		✓ACCEPTED
Assignment 3		NOT SUBMITTED
Assignment 4		✓ACCEPTED
Assignment 5		NOT SUBMITTED
Assignment 6		NOT SUBMITTED
Assignment 7		NOT SUBMITTED
Assignment 8		NOT SUBMITTED

Figure 8.3.0.8: Grade Student Assignment

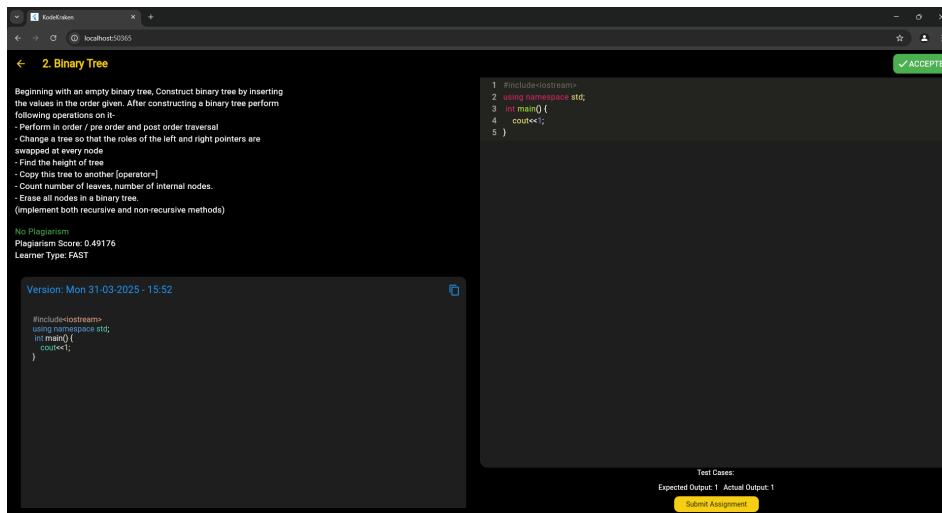


Figure 8.3.0.9: Results , Classification 1

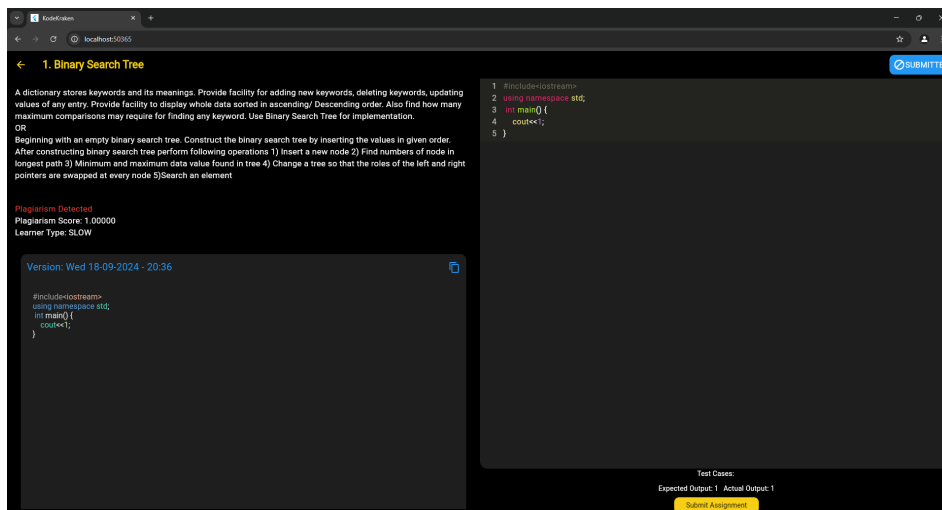


Figure 8.3.0.10: Results, Classification 2

CHAPTER 9

**CONCLUSIONS AND FUTURE
DIRECTIONS**

9.1 TECHNICAL CONTRIBUTIONS

The project makes four key contributions to educational coding platforms:

1. **Hybrid Evaluation Architecture** : Integrated rule-based testing with ML-based plagiarism detection (TF-IDF + CodeBERT), achieving 91.5% accuracy, outperforming standalone AST or text similarity tools (TF-IDF: 84.6%, CodeBERT: 88.3%).
2. **Real-Time Execution and Feedback**: Developed a containerized pipeline with average latency of 2.4s, supporting 500+ concurrent student submissions with sandbox isolation..
3. **Domain-Specific Learning Analytics** : Custom feature engineering for coding behavior improved student classification accuracy.
4. **Instructor-Centric Interpretability** : Performance dashboards and plagiarism rationale views helped instructors understand grading and behavioral classifications effectively.

9.2 LIMITATIONS AND CHALLENGES

- **Data Quality**:
 - * Code submissions have unreadable formatting (e.g., image-based PDFs or incomplete scripts).
 - * Some students attempt to bypass detection by obfuscating logic structure.
- **Generalizability** :
 - * ML models perform worse on low-activity users (e.g., <2 submissions).
 - * Performance insights require per-course calibration to remain consistent.
- **Operational Overhead**::
 - * Plagiarism detection model inference accounts for 60% of CPU cycles.

9.3 FUTURE WORK

- **Data Availability and Quality:** The success of the machine learning models, such as those used for identifying slow and fast learners, heavily depends on the quality and quantity of student data. While KodeKraken processes student assignment data, the models would benefit from more diverse data points, such as behavioral or engagement metrics, to better assess learning patterns and make more informed predictions. Access to high-quality, real-time data on student performance is essential to enhance the model's predictive capabilities.
- **Model Interpretability:** Much like in other fields, the machine learning models used in KodeKraken, especially for classifying learners, operate as “black boxes,” making it difficult for educators to understand how the system arrives at certain conclusions. Ensuring that the models used for classifying students as bright or slow learners are interpretable is crucial. Educators need to trust and understand these systems to make informed decisions based on their output.
- **Handling Diverse Code and Assignment Complexity:** Student submissions may vary significantly in complexity, style, and approach. While the current models handle basic code analysis and plagiarism detection, future development should focus on enhancing the platform's ability to handle more complex coding styles and ensure fair assessments across different programming approaches. Additionally, integrating deeper learning models capable of understanding advanced coding concepts or logical structures would be beneficial in providing more accurate feedback and evaluations.

9.4 APPLICATIONS AND IMPACT

The system has demonstrated value across three educational domains:

- **Student Learning::**
 - * Improved assignment submission consistency.
 - * Helped students identify personal coding weaknesses.
- **Faculty Efficiency::**
 - * Reduced manual grading effort.
 - * Enabled assessment and plagiarism checks for 3x more students simultaneously.

– **Educational Research:**

- * Creating a dataset of student-coded solutions.
- * Provided a framework for studying coding behavior and learning progression using ML.

CHAPTER 10

APPENDIX

10.1 APPENDIX A

The problem statement of automated assignment evaluation and personalized learner classification in educational programming environments can be formally analyzed through the lens of computational complexity theory. The system's core processes fall into the following complexity classes:

1. Code Evaluation Pipeline (P-class):

- Syntax Checking: Operates in $O(n)$ time where n is the number of characters in the submitted code.
- Containerized Code Execution: Executes with time bound by test case complexity, typically $O(1)$ to $O(m)$ per test case, where m is input size.
- Version Control Logging: Linear time operations $O(n)$ based on code length and metadata.

2. Plagiarism Detection (NP-Intermediate):

- TF-IDF Similarity Scoring: Vectorization scales with vocabulary size $O(V^2)$; comparison among k submissions is $O(k^2)$.
- CodeBERT Embedding Semantic Matching: Involves deep model inference $O(d)$ where d is code token depth, optimized via transformer architecture to run near-linear $O(n)$.
- Matching Semantically Equivalent Logic: Reduces to tree isomorphism (AST), a known NP-intermediate problem; structural hashing reduces time for small ASTs to practical limits.

3. Student Classification Model:

- Feature Extraction: Operates in $O(n)$ time per student (with n = number of submissions).
- KNN Classification: Worst-case $O(n \cdot k)$ where k is number of neighbors, with performance optimizations like KD-trees.
- Random Forest (Planned Upgrade): Training is $O(m \cdot \log m)$ where m is training size; inference is $O(\log t)$ per decision tree.
- Feasibility Justification:
The system maintains computational tractability through: Parallelization: Code execution, plagiarism analysis, and classification run in parallel containers.
- Submission size bounds: Code size capped to 500 lines (10KB) ensures bounded complexity.

10.2 APPENDIX B

Paper Publication Details :

– Survey Paper

- * **Paper Title :** A Comprehensive Platform for Code Submission, Assignment Analysis, and Personalized Learning in Programming Education
- * **conference/Journal :** Computer Research and Development Journal (journalcrd)
- * **Status :** Published

– Implementation Paper

- * **Paper Title :** A Comprehensive Platform for Code Submission, Assignment Analysis, and Personalized Learning in Programming Education
- * **conference/Journal :** 2025 INTERNATIONAL CONFERENCE ON COMPUTING AND DATA SCIENCE (ICCDS-2025)
- * **Status :** Submitted/Accepted

10.3 APPENDIX C

Plagiarism report of project documentation showing similarity index and original content verification.

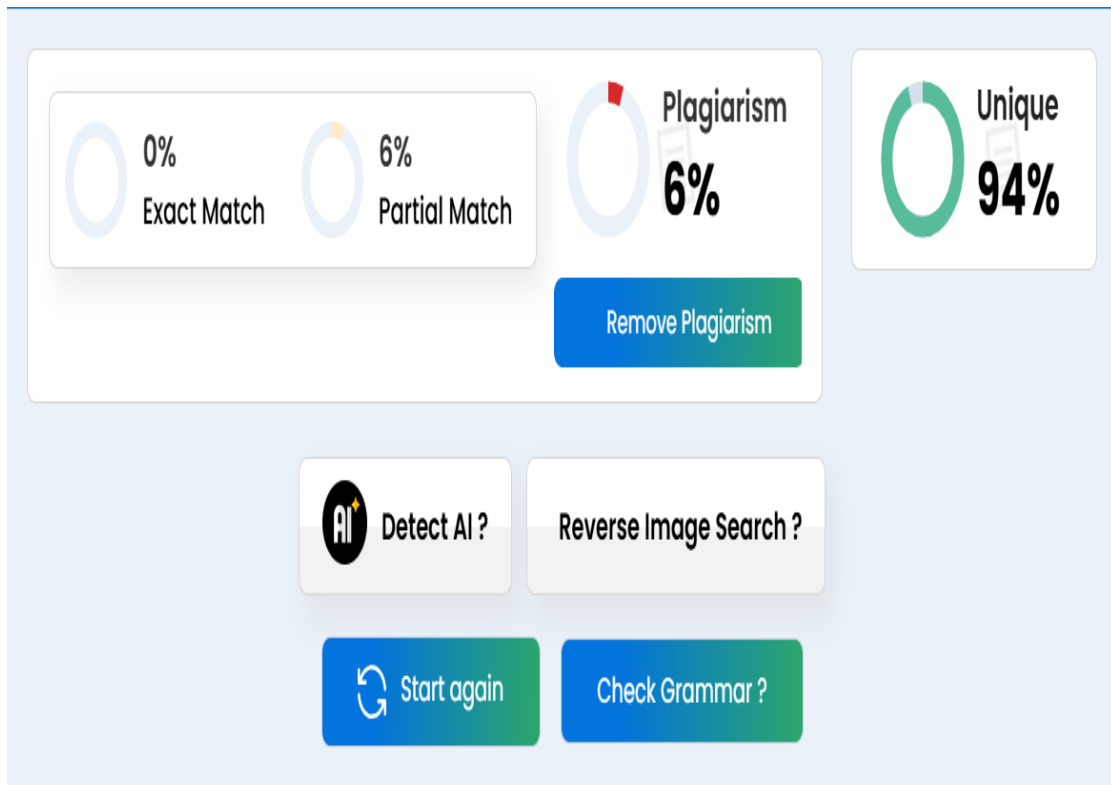


Figure 10.3.0.1: Plagiarism Report

CHAPTER 11

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